

# RUTHVIK BOMMENA

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## EDUCATION

### University of Illinois Urbana-Champaign (UIUC)

*Ph.D. Aerospace Engineering*

Expected May 2027

*M.Sc. Aerospace Engineering, Dean's List*

Aug 2021 – May 2023

### University of Illinois Chicago (UIC)

*B.Sc. Mechanical Engineering, Cum Laude & Dean's List*

Aug 2017 – May 2021

### Institute of Aviation at Parkland College

*Private Pilot License*

Aug 2022 – May 2023

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## EXPERIENCE

### Graduate Research Assistant, Department of Aerospace Engineering UIUC, Champaign, Illinois

Aug 2022 – Present

- Developing technologies for in-space servicing and assembly missions to enhance space sustainability, including developing path constrained trajectory optimization algorithms for multi-agent satellite systems
- Played a pivotal role in a collaborative project with NASA Jet Propulsion Laboratory and Ten One Aerospace on "Assembly of Large Aperture Space Telescopes in Cislunar Space Using a Swarm of Autonomous Small Satellites", as part of a NASA Phase 1 STTR project
- Responsible for comprehensive documentation and communication of project developments, including authoring formal reports for project deliverables and conducting bi-weekly technical presentations to update stakeholders on project progress

### Customer Success and Technical Support Specialist, SHIELD Illinois, Champaign, Illinois

Nov 2021 – June 2023

- Provided exceptional customer support at SHIELD Illinois, a nonprofit COVID-19 testing organization established by the University of Illinois System during the global pandemic to offer rapid, accurate, and affordable testing
- Efficiently addressed and resolved urgent issues and feedback from patients, clients, and partners, managing support tickets and using Point-and-Click (PNC) software for specimen tracking and client updates
- Collaborated effectively with a diverse, multi-cultural team to gather and analyze feedback for service improvement, leading to the identification and implementation of significant updates to the knowledge-base, dedicating extra hours to enhance customer experience

### Graduate Teaching Assistant, Department of Aerospace Engineering UIUC, Champaign, Illinois

Aug 2022 – Dec 2022

- Served as a Graduate Teaching Assistant for the AE 402 – Orbital Mechanics course, playing a key role in facilitating student understanding and engagement with complex subject matter
- Conducted weekly office hours, offering personalized assistance to students in comprehending lecture content, homework, and exam preparation, alongside the responsibility of grading various assignments to ensure timely and constructive feedback
- Took on the responsibility of preparing and delivering comprehensive course material during lectures as required, demonstrating adaptability and a thorough grasp of course concepts

### Mechanical Design Intern, Skyroot Aerospace, Hyderabad, India

May 2021 – Aug 2021

- Engineered a sophisticated four degree-of-freedom ring-type gimbal mount specifically for thrust vector control applications, showcasing a keen understanding of structural and propulsion systems
- Conducted comprehensive material analysis through Finite Element Analysis (FEA) to certify the gimbal's resilience under various structural and thermal loads, ensuring optimal performance and durability
- Developed expertise in thrust vectoring systems, honing skills in computer-aided design (CAD) and FEA software, complemented by proficiency in technical report writing and delivering impactful presentations, enhancing project documentation and stakeholder communication

- Engineering Intern, Defence Research and Development Organisation, Hyderabad, IndiaMay 2019 – July 2019
- Contributed to the fabrication process of the Naval Anti-Ship Missile–Short Range (NASM-SR) project at the prestigious Research Centre Imarat Dr. APJ Abdul Kalam Missile Complex, demonstrating proficiency in advanced manufacturing techniques and project execution
  - Successfully engineered an innovative integration method for different sections of the missile using optimal joint types, ensuring the design maintained consistent aerodynamic behavior and met operational specifications
  - Advanced my technical knowledge in the field of guided missiles, specializing in the modeling of missile systems and non-destructive testing for aerospace applications, thereby enhancing quality assurance and system reliability

UNDERGRADUATE PROJECTS

- Mechanical Design and Analysis Lead, UIC Capstone Senior Design Project, Chicago, IllinoisAug 2020 – May 2021
- Collaborated with senior design team members to develop a craft capable of carrying a high-altitude balloon to the edge of space, equipped with onboard cameras to film the entire journey from launch to landing
  - Skillfully assembled the craft by integrating multiple subsystems, including custom-made antennas, an electronics payload with video cameras, Global Positioning System (GPS), and an automatic landing system
  - Engineered a safe and reliable low-power release mechanism using a one-dimensional servo motor, designed to detach the payload from the balloon and enable its glide back to Earth for recovery

- Sub-Lead: Structures, American Institute of Aeronautics and Astronautics at UIC, Chicago, IllinoisSep 2017 – Jan 2020
- Worked with the structures team to analyze the structural dynamics of rockets under diverse loads, leading efforts in fabricating and integrating key components such as the nosecone, avionics bay, body, and motor assembly
  - Achieved first place in Student Research and Development (SRAD) 30,000ft and second place in Commercial off the Shelf (COTS) categories at the 2018 Intercollegiate Rocket Engineering Competition, contributing significantly to a team competing in high-powered rocket contests nationwide
  - Contributed to continuous improvement in rocket design and performance, employing advanced techniques in simulation and physical testing to optimize stability, durability, and efficiency under challenging flight conditions

PUBLICATIONS

- R. Bommena, and R. Woollands. "Novel Path Constrained Trajectory Optimization for Multi-Agent Systems." Submitted to *The Journal of the Astronautical Sciences* (2024)
- A. Pascarella, R. Bommena, S. Eggl, and R. Woollands. "Mission Design for Space Telescope Servicing at Sun-Earth L2." Submitted to *Acta Astronautica* (2024)

COMMUNITY SERVICE

- Student Volunteer, Shalivahana Foundation, Hyderabad, IndiaNov 2013 – July 2016
- Led an initiative to identify and support over 100 migrant child laborers at various construction sites, successfully facilitating their transition to safer environments and access to education
  - Demonstrated a strong commitment to social responsibility through proactive engagement with the NGO, contributing to the betterment of child welfare and community development
  - Recognized as a self-starter and motivated team player with an excellent work ethic, strong sense of duty, and deep understanding of social issues

SKILLS & RELEVANT COURSEWORK

- Software:** MATLAB & Simulink, Python, SolidWorks, ANSYS, MS Office Suite, XFOIL, LaTeX
- Relevant Coursework:** Optimal Spacecraft Trajectories, Orbital Mechanics, Learn/Control Multi-Agent Systems, Spacecraft Attitude Control, Optimum Control Systems, Planetary Entry, Systems Dynamics & Control, Numerical Methods, Mechanical Vibrations, Strength of Materials, Fluid Mechanics, Thermodynamics

LICENSES & CERTIFICATIONS

- Private Pilot License, Federal Aviation AdministrationMay 2023
- Spaceflight Engineering Certificate, UIUCMay 2023